INCH-POUND

MIL-T-46590C
30 April 1993
SUPERSEDING
MIL-T-46590B
28 January 1975

MILITARY SPECIFICATION

THRUSTER, CARTRIDGE ACTUATED, M15 METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M15 Cartridge Actuated Thruster Metal Parts Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-2550	Ammunition, General Specification For
MIL-W-8604	Welding of Aluminum Alloys: Process for

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-413	Visual Inspection Guide for Elastomeric O-rings

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.
(Unless otherwise indicated, copies of federal and military specifications and standards are available from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:23Z
Check the source to verify that this is the current version before use.

2.1.2 Other Government documents, drawings, and publications. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

Naval Surface Warfare Center, Indian Head (CAGE Codes 19200 and 19203)

10520876	Thruster, Cartridge Actuated, M15 Metal Parts Assembly
8858440	Box, Packing, Ammunition, For Thruster, Cartridge Actuated, M15
8858441	Box, Fiberboard (Inner), Packing, Ammunition, For Thruster, Cartridge Actuated, M15
9217560	Box, Fiberboard (Outer), Alternative, Packing Ammunition, for Thruster, Cartridge Actuated, M15

U.S. Air Force

TPO 00-607-0314	Thruster, Cartridge Actuated, M15

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Data Control Branch (Code 8410), Indian Head, MD 20640-5035.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to the first article inspection (see 4.3 and 6.3).

3.2 Materials. All materials used in the manufacture of the thrusters shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2.1 Raw material. The contractor shall be responsible for selecting from the classification list contained in the raw material specification the surface and physical condition of the raw material intended for use in manufacture.

3.3 Hardness and other physical properties. Hardness and other physical properties required by the drawing shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:23Z
Check the source to verify that this is the current version before use.

3.4 Hydrostatic pressure. The body shall withstand a hydrostatic pressure of 11,000 pounds per square inch (psi) for 15 seconds, minimum, without leaking, deformation, or mechanical failure (see 6.4.1).

3.5 Shear pin strength.

3.5.1 Shear pin. The shear pin, after application of protective finish, shall shear when a load of 55 ± 5 pounds is applied at the steady rate of 0.125 inch per minute.

3.5.2 Unlock shear pin. The unlock shear pin, after application of protective finish, shall shear when a load of 1150 ± 150 pounds is applied at the steady rate of 0.125 inch per minute.

3.6 Welding.

3.6.1 Reservoir assembly. The welding of the reservoir assembly shall comply with the requirements and shall be welded in accordance with MIL-W-8604 with the exceptions specified herein.

3.6.2 Internal pressure. The reservoir assembly shall withstand a constant internal air pressure of 90 pounds per square inch (psi), minimum, for 15 seconds, minimum, without application of any external load and shall show no evidence of leakage or mechanical failure (see 6.4.1).

3.6.3 Torque. The weld shall withstand a torque application of 10 inch pounds, minimum, without any breakage or deformation of any welded joint or component.

3.7 Metal parts assembly. Parts shown in table I shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing.

TABLE I. Assembly parts.

Assembled Parts	Method of Inspection
Rod end to piston rod	Hand tight
Firing pin housing to body	Hand tight
All setscrews and cap screws	Wrench tight

3.7.1 Piston rod rotation. Prior to final assembly of the ring nut, the piston rod shall be free to rotate in a clockwise direction for 360 degrees.

3.7.2 Torque.

3.7.2.1 Hollow lock screw. A torque of 5 inch pounds, maximum, shall be applied to seat the hollow lock screw on the gasket.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:23Z
Check the source to verify that this is the current version before use.

3.7.2.2 Ring nut. Torque requirements shall be as specified in table II. No movement or deformation shall occur upon application of the breakaway torque.

TABLE II. Torque requirements.

Components	A	B	C
	Run-on, minimum (inch pounds)	Seating or Bottoming (including A) (inch pounds)	Breakaway, minimum (inch pounds)
Ring nut, after assembly of rod piston and spacer piston	40	600 ± 30	400

3.7.3 Piston rod retraction. The piston rod shall not retract into the body upon application of a static compression load of 25 ± 5 pounds between mounting points.

3.7.4 Fluid leakage. The thruster shall not show evidence of fluid leakage when tested in Positions I and II (see 3.7.4.1 and 3.7.4.2) for a minimum of 12 hours each prior to the installation of the reservoir assembly.

3.7.4.1 Position I. Thruster shall not show evidence of fluid leakage at the orifice hole when the thruster is horizontal and the orifice hole pointing down.

3.7.4.2 Position II. Thruster shall not show evidence of fluid leakage at the piston when the thruster is vertical with the piston pointing down.

3.8 Protective finish. Protective finish shall be in accordance with the applicable specifications and shall apply to all surfaces of the component regardless of contour configuration.

3.9 Residual magnetism. The metal parts assembly shall not deflect the indicator of the compass more than five degrees in either direction using procedures in 4.6.9.

3.10 Workmanship. Thrusters shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.10.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.10.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.10.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:23Z
Check the source to verify that this is the current version before use.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.3 Inspections. Inspections shall be in accordance with MIL-A-2550.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: room ambient (68 to 77•F)
b. Altitude: sea level
c. Humidity: 50 percent relative, maximum
d. Atmospheric pressure: 14.7 psi

4.4 First article inspection. At the beginning of regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of five metal parts assemblies, nine sets of unassembled parts, one set of unassembled parts without plating or final protective finish, and fifty shear pins from the same lot of shear pins tested by the contractor. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, must be the same as will be used for regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the first article sample shall be inspected for all requirements of the drawings and specifications. The Government reserves the right to require new first article samples until an acceptable sample is submitted (see 6.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:23Z
Check the source to verify that this is the current version before use.

4.5 Quality conformance inspection.

4.5.1 Lot.

4.5.1.1 Submission of product. Unless otherwise specified by the contracting activity, formation and identification of lots shall be in accordance with MIL-STD-105.

4.5.1.2 Size of lot. The production lot size shall be as specified in the contract.

4.5.2 Examination. One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects. Four power (4x) magnification may be used in performing visual examination for such defects as burr and foreign matter. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

4.5.2.1 Pin, Shear, Drawing 8593876, covering a detail of Drawing 10520876.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Diameter, max	Gage
102	Total length, min	Gage
103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual
105	Burr	
Visual		
106	Foreign matter	Visual
Minor		
201	Total length, max	Gage
202	Profile of ends	Visual
203	Surface finish improper	Visual

4.5.2.2 Pin, Firing, Drawing 8593878, covering a detail of Drawing 10520876.

Critical		
None defined		

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:23Z
Check the source to verify that this is the current version before use.

Major		
101	Concentricity of groove with body	Gage
102	Concentricity of striker with body	Gage
103	Perpendicularity of striker face with body	Gage
104	True position of pin hole	Gage
105	Diameter of body, min	Gage
106	Diameter of groove	Gage
107	Diameter of striker	Gage
108	Diameter of pin hole, max	Gage
109	Depth of pin hole, max	Gage
110	Length of striker, min	Gage
111	Profile of groove	Gage
112	Metal defective	Visual
113	Protective finish inadequate or defective	Visual
114	Surface finish of body improper	Visual
115	Burr or sharp corners	Visual
116	Foreign matter	Visual
Minor		
201	Diameter of body, max	Gage
202	Diameter of pin hole, min	Gage
203	Depth of pin hole, min	Gage
204	Total length	
Gage		
205	Length of striker, max	Gage
206	Location of pin hole	Gage
207	Width of groove	Gage
208	Location of groove	Gage
209	Depth of slot in base	Gage
210	Width of slot in base	Gage
211	Location of slot in base	Gage
212	Marking missing, incorrect, or illegible	Visual
213	Radius incorrect	Gage
214	Surface finish improper, except as otherwise classified	Visual

4.5.2.3 Gasket, Drawing 10520878, covering a detail of Drawing 10520876.

Critical
None defined

Major		
101	Outside diameter	Gage
102	Inside diameter	Gage
103	Thickness	
Gage		
104	Material defects, flow line or splits	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:23Z
Check the source to verify that this is the current version before use.

105	Foreign matter	Visual

Minor
None defined

4.5.2.4 Plug, Drawing 10520879, covering a detail of Drawing 10520876.

Critical
None defined

Major		
101	Outside diameter	Gage
102	Inside diameter	Gage
103	Thickness	
Gage		
104	Thickness of web	Gage
105	Metal defective	Visual
106	Protective finish inadequate or defective	Visual
107	Radius incorrect	Gage
108	Surface finish improper	Visual
109	Foreign matter	Visual

Minor
None defined

4.5.2.5 Pin, Unlock Shear, Drawing 10520882, covering a detail of Drawing 10520876.

Critical
None defined

Major		
101	Diameter, max	Gage
102	Metal defective	Visual
103	Protective finish inadequate or defective	Visual
104	Foreign matter	Visual
Minor		
201	Total length	
Gage		
202	Profile of ends	Gage
203	Surface finish improper	Visual

4.5.2.6 Ring, Retaining, Drawing 10520883, covering a detail of Drawing 10520876.

Critical
None defined

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:23Z
Check the source to verify that this is the current version before use.

Major		
101	Inside diameter	Gage
102	Thickness	
	Gage	
103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual
105	Burr	
	Visual	
Minor		
201	Opening between ends	Gage
202	Radius or chamfer incorrect	Gage
203	Foreign matter	Visual

4.5.2.7 Rod, End, Drawing 10520884, covering a detail of Drawing 10520876.

Critical		
None defined		
Major		
101	Pitch diameter of thread, max	Gage
102	Major diameter of thread, max	Gage
103	True position of hole	Gage
104	Metal defective	Visual
105	Protective finish inadequate or defective	Visual
106	Burr	
	Visual	
107	Foreign matter	Visual
Minor		
201	Pitch diameter of thread, max	Gage
202	Diameter of undercut, max	Gage
203	Diameter of hole	Gage
204	Diameter of body	Gage
205	Depth to base of undercut	Gage
206	Length from hole to end of flat	Gage
207	Width across flat	Gage
208	Width of undercut	Gage
209	Location of hole	Gage
210	Location of flat	Gage
211	Radius incorrect	Gage
212	Marking missing, incorrect, or illegible	Visual
213	Surface finish improper	Visual

4.5.2.8 Spacer, Reservoir, Drawing 10520886, covering a detail of Drawing 10520876.

Critical		
None defined		

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:23Z
Check the source to verify that this is the current version before use.

Major		
101	Diameter of body	Gage
102	Diameter of pilot	Gage
103	Diameter of hole	Gage
104	Length to pilot	Gage
105	Total length	
Gage		
106	Metal defective	Visual
107	Protective finish inadequate or defective	Visual
108	Surface finish improper	Visual
109	Burr or sharp corner	Visual
110	Foreign matter	Visual
Minor		
201	Radius or chamfer incorrect	Gage

4.5.2.9 Piston, Gas, Drawing 10520887, covering a detail of Drawing 10520876.

Critical
None defined

Major		
101	Concentricity of internal groove with bore	Gage
102	Concentricity of shoulder with bore	Gage
103	Concentricity of external grooves with shoulder	Gage
104	Diameter of bore, max	Gage
105	Diameter of internal groove, max	Gage
106	Diameter of external groove, min	Gage
107	Diameter of shoulders, min	Gage
108	Length across shoulders, min	Gage
109	Total length, min	Gage
110	Width of internal groove, max	Gage
111	Width of external grooves	Gage
112	Profile of internal groove	Gage
113	Profile of external grooves	Gage
114	Radius or chamfer incorrect	Gage
115	Metal defective	Gage
116	Protective finish inadequate or defective	Visual
117	Surface finish improper	Visual
118	Burr or sharp corners	Visual
119	Foreign matter	Visual
Minor		
201	Diameter of bore, min	Gage
202	Diameter of internal groove, min	Gage
203	Diameter of external groove, max	Gage
204	Diameter of shoulders, max	Gage
205	Length across shoulders, max	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:23Z
Check the source to verify that this is the current version before use.

206	Total length, max	Gage
207	Width of internal groove, min	Gage
208	Diameter of body	Gage
209	Depth to internal groove	Gage
210	Location of external grooves	Gage
211	Marking missing, incorrect, or illegible	Visual

4.5.2.10 Spacer, Piston, Drawing 10520888, covering a detail of Drawing 10520876.

Critical
None defined

Major

101	Concentricity of internal groove with bore	Gage
102	Concentricity of external groove with shoulder	Gage
103	Concentricity of flange with shoulder	Gage
104	Concentricity of shoulder with bore	Gage
105	True position of hole	Gage
106	Diameter of bore, max	Gage
107	Diameter of internal groove, max	Gage
108	Diameter of flange, min	Gage
109	Diameter of external groove, min	Gage
110	Diameter of shear pin hole, max	Gage
111	Diameter of shoulder, min	Gage
112	Width of flange, min	Gage
113	Width of internal groove, max	Gage
114	Profile of internal groove	Gage
115	Profile of external groove	Gage
116	Radius or chamfer incorrect	Gage
117	Metal defective	Visual
118	Protective finish inadequate or defective	Visual
119	Surface finish improper	Visual
120	Burr or sharp corners	Visual
121	Foreign matter	Visual

Minor

201	Diameter of bore, min	Gage
202	Diameter of internal groove, min	Gage
203	Diameter of body	Gage
204	Diameter of flange, max	Gage
205	Diameter of external groove, max	Gage
206	Diameter of shear pin hole, min	Gage
207	Diameter of shoulder, max	Gage
208	Width of flange, max	Gage
209	Width of internal groove, min	Gage
210	Location of hole	Gage
211	Location of external groove	Gage
212	Depth to internal groove	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:23Z
Check the source to verify that this is the current version before use.

213	Length from shoulder to face of flange	Gage
214	Total length	Gage
215	Width of external groove	Gage
216	Marking missing, incorrect, or illegible	Visual

4.5.2.11 Body, Drawing 10520889, covering a detail of Drawing 10520876.

Critical
None defined

Major

101	Pitch diameter of thread, large cavity, max	Gage
102	Minor diameter of thread, large cavity, max	Gage
103	Pitch diameter of thread, small cavity, max	Gage
104	Minor diameter of thread, small cavity, max	Gage
105	Concentricity of outlet cavity with counterbore	Gage
106	Concentricity of thread with large cavity	Gage
107	Concentricity of body bore with large cavity	Gage
108	Concentricity of outlet hole with counterbore	Gage
109	Concentricity of thread with small cavity	Gage
110	Concentricity of counterbore with small cavity	Gage
111	Concentricity of setscrew thread with counterbore	Gage
112	Perpendicularity of boss faces with hole	Gage
113	True position of mounting holes, lug end	Gage
114	True position of mounting holes, threaded end	Gage
115	True position of small cavity with large cavity	Gage
116	True position of gas port	Gage
117	True position of counterbore, relief hole	Gage
118	True position of lug hole	Gage
119	Diameter of large cavity, max	Gage
120	Diameter of small cavity, max	Gage
121	Diameter of counterbore, small cavity, max	Gage
122	Diameter of undercut, small cavity, max	Gage
123	Diameter of undercut, large cavity, max	Gage
124	Diameter of undercut, base of large cavity, max, if applicable	Gage
125	Diameter of pressure groove, max	Gage
126	Diameter of hole in lug	Gage
127	Diameter of outlet hole	Gage
128	Diameter of small counterbore, outlet	Gage
129	Diameter of large counterbore, outlet	Gage
130	Depth of screw holes, threaded end, max	Gage
131	Depth to bottom of counterbore, small cavity	Gage
132	Depth to rear angle of pressure groove, max	Gage
133	Depth to front angle of pressure groove, max	Gage
134	Length of effective thread, screw holes, lug end, min	Gage
135	Length of effective thread, screw holes, threaded end, min	Gage
136	Length of effective thread, outlet hole, min	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:23Z
Check the source to verify that this is the current version before use.

137	Length between screw holes	Gage
138	Location of screw holes, threaded end	Gage
139	Location of hole in lug	Gage
140	Radius at mouth of small cavity	Gage
141	Metal defective	Visual
142	Protective finish inadequate or defective	Visual
143	Surface finish of large cavity and outlet counterbore improper	Visual
144	Burr or sharp corner	Visual
145	Foreign matter	Visual
146	Radius on lip at hole between cartridge cavity and gas cavity	Manual/Visual
Minor		
201	Pitch diameter of thread, large cavity, min	Gage
202	Pitch diameter of thread, small cavity, min	Gage
203	Pitch diameter of setscrew thread	Gage
204	Minor diameter of setscrew thread, max	Gage
205	Pitch diameter of thread, mounting holes	Gage
206	Minor diameter of thread, mounting holes, max	Gage
207	Diameter of large cavity, min	Gage
208	Diameter of small cavity, min	Gage
209	Diameter of counterbore, small cavity, min	Gage
210	Diameter of pressure groove, min	Gage
211	Diameter of gas port	Gage
212	Diameter of outlet cavity	Gage
213	Depth to base of large cavity	Gage
214	Depth of small counterbore, outlet hole	Gage
215	Depth of large counterbore, outlet hole	Gage
216	Depth of setscrew cavity	Gage
217	Depth of outlet cavity	Gage
218	Depth to rear angle of pressure groove, min	Gage
219	Depth to front angle of pressure groove, max	Gage
220	Width across lug	Gage
221	Depth to drill point, large cavity	Gage
222	Depth of screw holes, lug end, max	Gage
223	Diameter of body, threaded end	Gage
224	Diameter of body, lug end	Gage
225	Width of undercut, small cavity	Gage
226	Width of undercut, large cavity	Gage
227	Width of undercut base of large cavity, if applicable	Gage
228	Diameter of body, lug end	Gage
229	Diameter of pressure cavity	Gage
230	Diameter of body, offset	Gage
231	Depth to base of undercut, large thread	Gage
232	Depth to base of small cavity	Gage
233	Depth to base of pressure cavity	Gage
234	Depth to drill point, small cavity, max	Gage
235	Depth to base of undercut, small cavity	Gage
236	Depth from lug hole to body	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:23Z
Check the source to verify that this is the current version before use.

237	Length of body diameter, lug end, min	Gage
238	Profile of body exterior	Gage
239	Marking missing, incorrect, or illegible	Visual
240	Radius or chamfer incorrect, except as otherwise classified	Gage
241	Surface finish improper, except as otherwise classified	Visual

4.5.2.12 Plug, Reservoir, Drawing 10520891, covering a detail of Drawing 10520890.

Critical
None defined

Major

101	Metal defective	Visual

Minor

201	Length of large diameter	Gage
202	Large diameter	Gage
203	Thickness	Gage
204	Chamfer incorrect	Gage
205	Surface finish improper	Visual
206	Burr	Visual

4.5.2.13 Angle, Drawing 10520892, covering a detail of Drawing 10520890.

Critical
None defined

Major

101	Metal defective	Visual

Minor

201	Length	Gage
202	Width	Gage
203	Thickness	Gage
204	Height	Gage
205	Radius incorrect	Gage
206	Surface finish improper	Visual

4.5.2.14 Body, Reservoir, Drawing 10520893, covering a detail of Drawing 10520890.

Critical
None defined

Major

101	Diameter of tube, outside	Gage
102	Diameter of small hole	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:23Z
Check the source to verify that this is the current version before use.

103	Thickness of wall	Gage
104	Metal defective	Visual

Minor

201	Total length	Gage
202	Location of small hole	Gage
203	Surface finish improper	Visual

4.5.2.15 Inlet, Reservoir, Drawing 10520894, covering a detail of Drawing 10520890.

Critical
None defined

Major

101	Concentricity of center hole with counterbore	Gage
102	True position of radius at center hole	Gage
103	True position of counterbore at center hole	Gage
104	True position of holes at ends	Gage
105	Center line of segments in degrees	Gage
106	Diameter of counterbore, center hole	Gage
107	Diameter of center hole	Gage
108	Inside radius	Gage
109	Depth to bottom of counterbore, center hole	Gage
110	Metal defective	Visual
111	Surface finish improper	Visual

Minor

201	Diameter of clearance cut, center hole	Gage
202	Depth of clearance cut, center hole	Gage
203	Diameter of counterbores, end holes	Gage
204	Depth of counterbores, end holes	Gage
205	Thickness	Gage
206	Location of end holes	Gage
207	Burr or sharp corners	Visual
208	Foreign matter	

4.5.2.16 Strap, Drawing 10520899, covering a detail of Drawing 10520890.

Critical
None defined

Major

101	True position of holes	Gage
102	Metal defective	Visual

Minor

201	Diameter of holes	Gage
202	Length	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:23Z
Check the source to verify that this is the current version before use.

203	Width	Gage
204	Location of holes	Gage
205	Thickness	Gage
206	Inside radius	Gage
207	Surface finish improper	Gage

4.5.2.17 Firing Pin Housing and Pellet Assembly, Drawing 10520880, covering a detail of Drawing 10520876.

Critical
None defined

Major

101	Pitch diameter of external thread, min	Gage
102	Major diameter of external thread, min	Gage
103	Protrusion of pellets	Gage
104	Pitch diameter of internal thread, max	Gage
105	Minor diameter of internal thread, max	Gage
106	Concentricity of long cavity with external thread	Gage
107	Concentricity of boss with external cavity	Gage
108	Perpendicularity of boss face with cavity	Gage
109	True position of pin hole	Gage
110	Diameter of internal thread pilot	Gage
111	Diameter of long cavity, max	Gage
112	Diameter of pin hole, max	Gage
113	Diameter of boss	Gage
114	Length of effective internal thread, min	Gage
115	Length of internal thread, max	Gage
116	Height of boss	Gage
117	Thread damaged or not full	Visual
118	Metal defective	Visual
119	Protective finish inadequate or defective	Visual
120	Surface finish in cavity or internal threaded end improper	Visual
121	Burr or sharp edge	Visual
122	Foreign matter	Visual
123	Pellet missing, loose, or improperly inserted	Manual/Visual
124	Thread damaged	Visual
125	Burr	Visual
126	Foreign matter	Visual

Minor

201	Location of pellets	Gage
202	Pitch diameter of external thread, max	Gage
203	Pitch diameter of internal thread, min	Gage
204	Pitch diameter of setscrew thread	Gage
205	Minor diameter of setscrew thread	Gage
206	Diameter of long cavity, min	Gage
207	Diameter of pin hole, min	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:23Z
Check the source to verify that this is the current version before use.

208	Diameter of counterbore, threaded cavity	Gage
209	Diameter of threaded cavity	Gage
210	Diameter of bore	Gage
211	Depth of threaded cavity	Gage
212	Depth of setscrew cavity, max	Gage
213	Depth of counterbore, threaded cavity	Gage
214	Depth of long cavity	Gage
215	Total length	Gage
216	Length of effective thread, setscrew cavity, min	Gage
217	Location of pin hole	Gage
218	Diameter of chamfer, threaded cavity	Gage
219	Diameter of body	Gage
220	Length of flat	Gage
221	Length of external thread	Gage
222	Width across flat	Gage
223	Location of flat	Gage
224	Marking missing, incorrect, or illegible	Visual
225	Radius or chamfer incorrect	Gage
226	Surface finish improper, except as otherwise classified	Visual
227	Pellet radius incorrect	Visual

4.5.2.18 Ring Nut and Pellet Assembly, Drawing 10520881, covering a detail of Drawing 10520876.

Critical
None defined

Major

101	Pitch diameter of thread, min	Gage
102	Major diameter of thread, min	Gage
103	Concentricity of basic diameter with bore	Gage
104	Concentricity of bore with pitch diameter of thread	Gage
105	Concentricity of small counterbore with bore	Gage
106	Basic diameter	Gage
107	Diameter of small counterbore, min	Gage
108	Diameter of bore, max	Gage
109	Depth of large counterbore, min	Gage
110	Length to basic diameter	Gage
111	Angle at basic diameter	Gage
112	Location of pellets	Gage
113	Thread damaged or not full	Visual
114	Metal defective	Visual
115	Protective finish inadequate or defective	Visual
116	Surface finish improper	Visual
117	Protrusion of pellet	Gage
118	Pellet missing, loose, or improperly inserted	Manual/Visual
119	Thread damaged	Visual
120	Burr	Visual
121	Foreign matter	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:23Z
Check the source to verify that this is the current version before use.

Minor

201	Pitch diameter of thread, max	Gage
202	Diameter of bore, min	Gage
203	Diameter of wrench holes	Gage
204	Diameter of small counterbore, max	Gage
205	Diameter of bolt circle	Gage
206	Depth of large counterbore, max	Gage
207	Diameter of large counterbore	Gage
208	Depth of wrench holes	Gage
209	Length of chamfer, small counterbore	Gage
210	Location of wrench holes	Gage
211	Thickness	Gage
212	Radius or chamfer incorrect, except as otherwise classified	Gage
213	Location of pellets	Gage
214	Pellet radius incorrect	Visual
215	Marking missing, incorrect, or illegible	Visual

4.5.2.19 Piston Rod Assembly, Drawing 10520885, covering a detail of Drawing 10520876.

Critical
None defined

Major

101	Pitch diameter of thread, min	Gage
102	Major diameter of thread, min	Gage
103	True position of pitch diameter of thread with large body	Gage
104	True position of small body with large body	Gage
105	True position of pin hole with large body	Gage
106	Diameter of large body, min	Gage
107	Diameter of pin hole in body, max	Gage
108	Diameter of small body, min	Gage
109	Length to pin hole	Gage
110	Length to small body	Gage
111	Total length	Gage
112	Length of effective thread, min	Gage
113	Length of thread, max	Gage
114	Thread damaged or not full	Visual
115	Sharp corner on body missing	Visual
116	Number of relief grooves incorrect	Visual
117	Radius or chamfer incorrect	Gage
118	Diameter over top of pellets	Gage
119	Metal defective	Visual
120	Protective finish inadequate or defective	Visual
121	Surface finish on body improper	Visual
122	Burr	Visual
123	Foreign matter	Visual
124	Pellet missing, loose, or improperly inserted	Manual/Visual
125	Thread damaged	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:23Z
Check the source to verify that this is the current version before use.

Minor		
201	Pitch diameter of thread, max	Gage
202	Diameter of large body, max	Gage
203	Diameter of pin hole in body, min	Gage
204	Diameter of small body, max	Gage
205	Thickness of head	Gage
206	Diameter of head	Gage
207	Depth of pin hole	Gage
208	Width of relief grooves in head	Gage
209	Location of pellets	Gage
210	Pellet radius incorrect	Gage
211	Marking missing, incorrect, or illegible	Gage
212	Surface finish improper, except as otherwise classified	Visual

4.5.2.20 Reservoir Assembly, Drawing 10520890, covering a detail of Drawing 10520876.

Critical
None defined

Major		
101	True position of mounting holes	Gage
102	Length between mounting holes	Gage
103	Liquid in tube	Manual/Visual
104	Inlet hole obstructed	Visual
105	Improper assembly	Visual
106	Metal defective or damaged	Visual
107	Protective finish inadequate or defective	Visual
108	Burr	Visual
109	Foreign matter	Visual
Minor		
201	Incorrect weld size	Gage
202	Incorrect weld profile	Gage
203	Incomplete weld	Visual
204	Undercutting	Visual
205	Surface defects, slag, or beads	Visual
206	Marking missing, incorrect, or illegible	Visual

4.5.2.21 Spacer, Piston, prior to installation on Rod, consisting of components 10520888, AN6227-B13, AN6227-B20, MS35803-115, covering a detail of Drawing 10520876.

Critical
None defined

Major		
101	Improper assembly	Visual
102	Ring missing	Visual
103	Packing missing	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:23Z
Check the source to verify that this is the current version before use.

104	Material defective	Visual
105	Burr	Visual
106	Foreign matter	Visual

Minor
None defined

4.5.2.22 Piston, Gas, prior to installation on Rod, consisting of components 10520887, MS29513-122, MS35803-115, AN6227-B13, covering a detail of Drawing 10520876.

Critical
None defined

Major

101	Improper assembly	Visual
102	Ring missing	Visual
103	Packing missing	Visual
104	Material defective	Visual
105	Burr	Visual
106	Foreign matter	Visual

Minor
None defined

4.5.2.23 Piston Rod, Gas Piston, and Piston Spacer Subassembly, consisting of components 10520885, 10520887, 10520882, 10520883, 10520888, AN6227-B20, AN6227-B13, MS29513-122, MS35803-115, covering a detail of Drawing 10520876.

Critical
None defined

Major

101	Improper assembly	Visual
102	Any part missing	Visual
103	Material defective	Visual
104	Burr	Visual
105	Foreign matter	Visual

Minor
None defined

4.5.2.24 Metal Parts Assembly, Drawing 10520876, prior to installation of Hollow Screw, 10520900, and Reservoir Assembly, Drawing 10520890.

Critical

1	Gasket missing or incorrectly installed	Manual/Visual
2	Plug (frangible) missing or incorrectly installed	Manual/Visual
3	Oil hole obstructed	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:23Z
Check the source to verify that this is the current version before use.

Major		
101	Any part missing, not otherwise classified	Visual
102	Improper assembly, not otherwise classified	Visual
103	Thread damaged	Visual
104	Foreign matter	Visual
Minor		
201	Protective finish inadequate or defective	Visual

4.5.2.25 Metal Parts Assembly, Drawing 10520876.

Critical		
1	Evidence of oil leakage	Visual
Major		
101	Crack in weld on reservoir	Visual
102	Excessive distortion of reservoir	Visual
103	Any part missing	Visual
104	Improper assembly	Visual
105	Marking missing, incorrect, or illegible	Visual
Minor		
201	Protective finish inadequate or defective	Visual
202	Foreign matter	Visual

4.5.2.26 Box, Fiberboard (Inner), Packing, Ammunition, Drawing 8858441.

Critical
None defined

Major
None defined

Minor		
201	Contents loose	Manual
202	Box damaged	Visual
203	Marking missing, incorrect, or illegible	Visual
204	Cut, damaged, or improperly sealed bag	Visual

4.5.2.27 Box, Packing, Ammunition, Drawing 8858440.

Critical
None defined

Major
None defined

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:23Z
Check the source to verify that this is the current version before use.

Minor		
201	Contents loose	Visual/Manual
202	Strapping missing, loose, broken, or improperly assembled	Visual/Manual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Number of units incorrect	Visual

4.5.2.28 Box, Fiberboard (Outer), Alternative, Packing, Ammunition, Drawing 9217560.

Critical
None defined

Major
None defined

Minor		
201	Contents loose	Manual/Visual
202	Carton broken or damaged	Visual
203	Number of units incorrect	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Tape incomplete	Visual

4.5.3 Inspection testing. The following tests shall be performed in accordance with 4.6.

4.5.3.1 Hardness. Fifty units from each heat treated lot of components shall be tested for compliance with the requirements on the applicable drawing. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot of components.

4.5.3.2 Hydrostatic pressure. Each body shall be subjected to the hydrostatic pressure test. Failure of any cap or chamber subassembly to comply with 3.4 shall be cause for rejection of the subassembly.

4.5.3.3 Shear pin strength. Fifty pins of each type from each lot of shear pins shall be tested for shear strength. Failure of any shear pin to comply with 3.5 shall be cause for rejection of the lot.

4.5.3.4 Protective finish. The number of units for protective finish tests shall be in accordance with the applicable specification. Failure of any unit in the inspection sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.5.3.5 Residual magnetism. Each assembly shall be subjected to the test for excessive residual magnetism as specified in 4.6.7. Failure of any unit in the inspection sample to comply with 3.9 shall be cause for rejection of the assembly.

4.5.3.6 Welding.

4.5.3.6.1 Reservoir assembly. The welding procedure and the welding equipment shall be subjected to tests for consideration of approval. The welding operator shall be subjected to a qualification test. Failure

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:23Z
Check the source to verify that this is the current version before use.

to comply with the requirement of the applicable specification shall be cause for rejection of the welded reservoir assemblies and all welded components (see 4.6.4.1).

4.5.3.6.1.1 Internal pressure. The reservoir assembly shall be subjected to an internal air pressure application. The number of units for test shall be as specified in MIL-STD-105. Failure to comply with the requirement of 3.6.2 shall be cause for rejection of the lot.

4.5.3.6.2 Torque. The welded joints shall be subjected to a torque application. The number of units for test shall be as specified in MIL-STD-105. Failure to comply with the requirements of 3.6.3 shall be cause for rejection of the lot.

4.5.3.7 Assembly.

4.5.3.7.1 Piston rod rotation. Each piston and body subassembly shall be subjected to a rotation of the piston rod within the body. Failure to comply with the requirement of 3.7.1 shall be cause for rejection of the subassembly.

4.5.3.7.2 Breakaway torque. Each body and ring nut subassembly shall be subjected to a breakaway torque test. Failure to comply with the requirements of table II (column C) shall be cause for rejection of the subassembly.

4.5.3.7.3 Piston rod retraction. Each piston rod and body subassembly shall be subjected to a static compression load application. Failure to comply with the requirement of 3.7.3 shall be cause for rejection of the subassembly.

4.5.3.7.4 Fluid leakage. Each thruster shall be subjected to leakage tests. Failure of any thruster to comply with the requirement of 3.7.4 shall be cause for rejection of the thruster.

4.6 Test methods and procedures.

4.6.1 Hardness. Prior to the application of protective finish, a hardness tester shall be used to determine compliance with the requirements of the applicable drawing.

4.6.2 Hydrostatic pressure. Prior to application of protective finish the body shall be mounted in a device, using a fixture for properly sealing all openings, and subjected to the internal hydrostatic pressure to determine compliance with 3.4.

4.6.3 Shear pin strength. After application of protective finish, the pin shall be subjected to the shear pin strength test to determine compliance with 3.5. The test shall be performed in a fixture in conjunction with a tensile testing machine.

4.6.4 Welding.

4.6.4.1 Reservoir assembly. Unless otherwise directed by the contracting officer, approval of the welding procedure, equipment, and the qualification of the welding officer shall be in accordance with the applicable specification, except that X-ray inspection and radiographic examination will not be required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:23Z
Check the source to verify that this is the current version before use.

4.6.4.2 Internal pressure. Prior to application of protective finish, the welded reservoir cylinder shall be mounted in a device and the pressure shall be applied as specified in 3.6.2.

4.6.4.3 Torque. A torque wrench, set at the requirement as specified in 3.6.3, shall be used in conjunction with a fixture to determine the adequacy of the joint. The torque shall be applied at each end of the reservoir cylinder with the load axis, moment arm axis, and long axis of the cylinder all mutually perpendicular. The reservoir assembly may be in the mounted position in the metal parts assembly or in any manner similar thereto.

4.6.5 Assembly.

4.6.5.1 Piston rod rotation. Prior to torque application on the ring nut, the piston rod shall be rotated manually, in a clockwise direction for 360 degrees.

4.6.5.2 Breakaway torque. A torque wrench, set at the requirement as specified in 3.7.2.2, shall be used in conjunction with a fixture to determine the adequacy of the joint. The torque shall be applied to the ring nut after insertion of pellets.

4.6.5.3 Piston rod retraction. After assembly of the piston rod into the body and the torque has been applied to the ring nut, the static compression load as specified in 3.7.3 shall be applied between the mounting points of the thruster.

4.6.5.4 Fluid leakage. The thruster shall be placed on paper or any material that will show oil or fluid markings. The thruster shall be positioned as noted in 3.7.4.1 and 3.7.4.2, and shall remain in each position for a minimum of 12 hours. The test shall be performed after the piston rod and ring nut have been assembled into the thruster, but prior to installation of the reservoir assembly.

4.6.6 Protective finish. Test procedures for protective finish shall be in accordance with the applicable specification referenced on the applicable drawing.

4.6.7 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end and at the same height as the compass, in a north-south horizontal position (head end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly for four full passes.

4.7 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with the requirements of section 5.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be Level A or B, as specified by the contracting activity (see 6.2). Level A shall be in accordance with Drawings 8858440 and 8858441. Level B shall be in accordance with Drawings 8858441 and 9217560.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:23Z
Check the source to verify that this is the current version before use.

5.2 Air Force packaging. Air Force packaging shall be in accordance with TPO 00-607-0314.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The thruster assembly is used in conjunction with other propellant actuated devices in Aircrew Emergency Escape Systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2)

c. If first article inspection is required (see 3.1 and 6.3)

d. The responsible inspection activity (see 4.1)

e. Any special instruction requirements (see 4.3)

f. Level of packing and marking (see 5.1)

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors that the items should be a first article sample. The first article should consist of 5 metal parts assemblies, 9 sets of unassembled parts, 1 set of unassembled parts without plating or final protective finish, and 50 shear pins. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definition.

6.4.1 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.5 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:23Z
Check the source to verify that this is the current version before use.

6.6 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

Custodian:

Preparing activity
Navy - OS

Navy - OS

Air Force - 99

(Project 1377-0E25)

Review activity:
Air Force - 70

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:23Z
Check the source to verify that this is the current version before use.